

$$\begin{aligned}
& y_u^{i1i2} + \frac{1}{16\pi^2} \left(\frac{1}{144} (18 y_u^{pi2} (\overline{y_d^{pr}} y_d^{i1r} + 3 \overline{y_u^{pr}} y_u^{i1r}) + y_u^{i1i2} (37 g_1^2 + 135 g_2^2 - 96 g_3^2)) - \frac{1}{36} g_1^2 \right. \\
& y_u^{i1i2} LF_{1,1,0}[m_1, m_q^{i1}] + \frac{1}{72} g_1^2 y_u^{i1i2} LF_{2,1,-1}[m_1, m_q^{i1}] - \frac{4}{9} g_1^2 y_u^{i1i2} LF_{1,1,0}[m_1, m_u^{i2}] + \\
& \frac{2}{9} g_1^2 y_u^{i1i2} LF_{2,1,-1}[m_1, m_u^{i2}] - \frac{3}{4} g_2^2 y_u^{i1i2} LF_{1,1,0}[m_2, m_q^{i1}] + \\
& \frac{3}{8} g_2^2 y_u^{i1i2} LF_{2,1,-1}[m_2, m_q^{i1}] - \frac{4}{3} g_3^2 y_u^{i1i2} LF_{1,1,0}[m_3, m_q^{i1}] + \\
& \frac{2}{3} g_3^2 y_u^{i1i2} LF_{2,1,-1}[m_3, m_q^{i1}] - \frac{4}{3} g_3^2 y_u^{i1i2} LF_{1,1,0}[m_3, m_u^{i2}] + \\
& \frac{2}{3} g_3^2 y_u^{i1i2} LF_{2,1,-1}[m_3, m_u^{i2}] - \frac{1}{2} \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{1,1,0}[m_d^r, \tilde{\mu}] + \\
& \frac{1}{2} \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} y_u^{i1i2} LF_{2,1,0}[m_l^p, m_e^r] - \frac{1}{2} \tilde{\mu}^2 \overline{y_e^{pr}} y_e^{pr} y_u^{i1i2} LF_{3,1,-1}[m_l^p, m_e^r] + \\
& \tilde{\mu} y_e^{pr} y_u^{i1i2} (C_{\phi1\phi2} \overline{a_e^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_e^{pr}}) LF_{3,1,0}[m_l^p, m_e^r] - \\
& 3 \tilde{\mu} y_e^{pr} y_u^{i1i2} (C_{\phi1\phi2} \overline{a_e^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_e^{pr}}) LF_{4,1,-1}[m_l^p, m_e^r] + \\
& 2 \tilde{\mu} y_e^{pr} y_u^{i1i2} (C_{\phi1\phi2} \overline{a_e^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_e^{pr}}) LF_{5,1,-2}[m_l^p, m_e^r] + \\
& \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} y_u^{i1i2} LF_{2,1,0}[m_q^p, m_d^r] - \frac{3}{2} \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{pr} y_u^{i1i2} LF_{3,1,-1}[m_q^p, m_d^r] + \\
& 3 \tilde{\mu} y_d^{pr} y_u^{i1i2} (C_{\phi1\phi2} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{3,1,0}[m_q^p, m_d^r] - \\
& 9 \tilde{\mu} y_d^{pr} y_u^{i1i2} (C_{\phi1\phi2} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{4,1,-1}[m_q^p, m_d^r] + \\
& 6 \tilde{\mu} y_d^{pr} y_u^{i1i2} (C_{\phi1\phi2} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{5,1,-2}[m_q^p, m_d^r] - \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{1,1,0}[m_q^p, \tilde{\mu}] + \\
& \frac{3}{2} y_u^{i1i2} \overline{a_u^{pr}} a_u^{pr} LF_{2,1,0}[m_u^r, m_q^p] - \frac{3}{2} y_u^{i1i2} \overline{a_u^{pr}} a_u^{pr} LF_{3,1,-1}[m_u^r, m_q^p] + \\
& 3 y_u^{i1i2} \overline{a_u^{pr}} (C_{\phi1^2} a_u^{pr} + C_{\phi1\phi2} \tilde{\mu} y_u^{pr}) LF_{3,1,0}[m_u^r, m_q^p] - \\
& 9 y_u^{i1i2} \overline{a_u^{pr}} (C_{\phi1^2} a_u^{pr} + C_{\phi1\phi2} \tilde{\mu} y_u^{pr}) LF_{4,1,-1}[m_u^r, m_q^p] + \\
& 6 y_u^{i1i2} \overline{a_u^{pr}} (C_{\phi1^2} a_u^{pr} + C_{\phi1\phi2} \tilde{\mu} y_u^{pr}) LF_{5,1,-2}[m_u^r, m_q^p] - \frac{1}{2} \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{1,1,0}[m_u^r, \tilde{\mu}] - \\
& \frac{3}{4} g_1^2 y_u^{i1i2} LF_{2,1,-1}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^2 y_u^{i1i2} LF_{3,1,-2}[\tilde{\mu}, m_1] - \\
& 2 C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{3,1,-1}[\tilde{\mu}, m_1] + m_1 C_{\phi1\phi2} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{3,1,0}[\tilde{\mu}, m_1] + \\
& 4 C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{4,1,-2}[\tilde{\mu}, m_1] - 3 m_1 C_{\phi1\phi2} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{4,1,-1}[\tilde{\mu}, m_1] - \\
& 2 C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{5,1,-3}[\tilde{\mu}, m_1] + 2 m_1 C_{\phi1\phi2} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{5,1,-2}[\tilde{\mu}, m_1] - \\
& \frac{9}{4} g_2^2 y_u^{i1i2} LF_{2,1,-1}[\tilde{\mu}, m_2] + \frac{3}{2} g_2^2 y_u^{i1i2} LF_{3,1,-2}[\tilde{\mu}, m_2] - \\
& 6 C_{\phi1^2} g_2^2 y_u^{i1i2} LF_{3,1,-1}[\tilde{\mu}, m_2] + 3 m_2 C_{\phi1\phi2} \tilde{\mu} g_2^2 y_u^{i1i2} LF_{3,1,0}[\tilde{\mu}, m_2] + \\
& 12 C_{\phi1^2} g_2^2 y_u^{i1i2} LF_{4,1,-2}[\tilde{\mu}, m_2] - 9 m_2 C_{\phi1\phi2} \tilde{\mu} g_2^2 y_u^{i1i2} LF_{4,1,-1}[\tilde{\mu}, m_2] - \\
& 6 C_{\phi1^2} g_2^2 y_u^{i1i2} LF_{5,1,-3}[\tilde{\mu}, m_2] + 6 m_2 C_{\phi1\phi2} \tilde{\mu} g_2^2 y_u^{i1i2} LF_{5,1,-2}[\tilde{\mu}, m_2] + \\
& \frac{1}{4} \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{2,1,-1}[\tilde{\mu}, m_d^r] + \frac{1}{2} \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{2,1,-1}[\tilde{\mu}, m_q^p] + \\
& \frac{1}{4} \overline{y_u^{pr}} y_u^{pi2} y_u^{i1r} LF_{2,1,-1}[\tilde{\mu}, m_u^r] + \frac{2}{9} m_1 g_1^2 a_u^{i1i2} LF_{1,1,1,0}[m_1, m_q^{i1}, m_u^{i2}] - \\
& \frac{1}{18} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{2,2,1,-1}[m_1, m_q^{i1}, m_u^{i2}] - \\
& \frac{1}{6} g_1^2 y_u^{i1i2} LF_{1,1,1,-1}[m_1, m_q^{i1}, \tilde{\mu}] - \frac{1}{18} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) \\
& LF_{2,2,1,-1}[m_1, m_u^{i2}, m_q^{i1}] + \frac{2}{3} g_1^2 y_u^{i1i2} LF_{1,1,1,-1}[m_1, m_u^{i2}, \tilde{\mu}] - \\
& \frac{1}{12} C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_q^{i1}] + \frac{1}{12} m_1 C_{\phi1\phi2} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_q^{i1}] + \\
& \frac{1}{3} C_{\phi1^2} g_1^2 y_u^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_u^{i2}] - \frac{1}{3} m_1 C_{\phi1\phi2} \tilde{\mu} g_1^2 y_u^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_u^{i2}] + \\
& \frac{3}{2} g_2^2 y_u^{i1i2} LF_{1,1,1,-1}[m_2, m_q^{i1}, \tilde{\mu}] + \frac{3}{4} C_{\phi1^2} g_2^2 y_u^{i1i2} LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_q^{i1}] - \\
& \frac{3}{4} m_2 C_{\phi1\phi2} \tilde{\mu} g_2^2 y_u^{i1i2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_q^{i1}] + \frac{8}{3} m_3 g_3^2 a_u^{i1i2} LF_{1,1,1,0}[m_3, m_q^{i1}, m_u^{i2}] - \\
& \frac{2}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{2,2,1,-1}[m_3, m_q^{i1}, m_u^{i2}] - \\
& \frac{2}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{2,2,1,-1}[m_3, m_u^{i2}, m_q^{i1}] + \\
& \tilde{\mu}^2 \overline{y_d^{pr}} y_d^{i1r} y_u^{pi2} LF_{1,1,1,0}[m_d^r, m_q^p, \tilde{\mu}] + \frac{1}{2} \tilde{\mu} y_d^{i1r} y_u^{pi2} (C_{\phi1\phi2} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) \\
& LF_{2,1,1,0}[m_d^r, m_q^p, \tilde{\mu}] - \frac{1}{2} \tilde{\mu} y_d^{i1r} y_u^{pi2} (C_{\phi1\phi2} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{3,1,1,-1}[m_d^r, m_q^p, \tilde{\mu}] - \\
& \frac{1}{4} \tilde{\mu} y_d^{i1r} y_u^{pi2} (C_{\phi1\phi2} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{2,2,1,-1}[m_d^r, \tilde{\mu}, m_q^p] + \\
& \frac{1}{2} \tilde{\mu} y_d^{i1r} y_u^{pi2} (C_{\phi1\phi2} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{2,1,1,0}[m_q^p, m_d^r, \tilde{\mu}] - \\
& \frac{1}{2} \tilde{\mu} y_d^{i1r} y_u^{pi2} (C_{\phi1\phi2} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{3,1,1,-1}[m_q^p, m_d^r, \tilde{\mu}] - \\
& \frac{1}{4} \tilde{\mu} y_d^{i1r} y_u^{pi2} (C_{\phi1\phi2} \overline{a_d^{pr}} + C_{\phi1^2} \tilde{\mu} \overline{y_d^{pr}}) LF_{2,2,1,-1}[m_q^p, \tilde{\mu}, m_d^r] + \\
& \frac{1}{9} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{2,1,1,0}[m_q^{i1}, m_1, m_u^{i2}] - \\
& \frac{1}{9} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{3,1,1,-1}[m_q^{i1}, m_1, m_u^{i2}] + \\
& \frac{4}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{2,1,1,0}[m_q^{i1}, m_3, m_u^{i2}] - \\
& \frac{4}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{3,1,1,-1}[m_q^{i1}, m_3, m_u^{i2}] + \\
& \frac{1}{9} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{2,1,1,0}[m_u^{i2}, m_1, m_q^{i1}] - \\
& \frac{1}{9} m_1 g_1^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{3,1,1,-1}[m_u^{i2}, m_1, m_q^{i1}] + \\
& \frac{4}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{2,1,1,0}[m_u^{i2}, m_3, m_q^{i1}] - \\
& \frac{4}{3} m_3 g_3^2 (C_{\phi1^2} a_u^{i1i2} + C_{\phi1\phi2} \tilde{\mu} y_u^{i1i2}) LF_{3,1,1,-1}[m_u^{i2}, m_3, m_q^{i1}] \Big)
\end{aligned}$$